JiaYu (Jasper) Hu

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EDUCATION

University of New South Wales | **Sydney, NSW**
Master of Commerce in Business Analytics | Feb. 2026 - Feb. 2027 (Expected)

Hong Kong Baptist University | **Hong Kong SAR**
Bachelor of Business Administration (Honours) in Accounting | Sep. 2021 - Jun. 2025
Completed at Beijing Normal-Hong Kong Baptist University (BNBU), Zhuhai campus

- *Second Class Honours*
- *Dean's List (2022-2023, 2023-2024, 2024-2025)*

RESEARCH INTERESTS

- LLM evaluation
- Retrieval-augmented generation
- Knowledge graphs
- ESG and sustainability disclosure

PUBLICATIONS & WORKING PAPERS

Ding, Y., Yang, Z., Lu, Y., Lan, L., Tang, Y., Yang, Z., Tang, X., **Hu, J.**, Ju, M., Zhang, W., Yang, W., Xu, X., & Zhang, W. *EulerESG: A unified platform for standards-aware ESG disclosure analysis*. Manuscript submitted to IEEE International Conference on Data Mining (ICDM 2026), Demo Track; under review.

RESEARCH EXPERIENCES

Reproducible Diagnostics and Benchmark Design for ESG Disclosure Assessment
UNSW, Sydney [Code] | Aug. 2026 - Present
Supervisor: Dr. Zhengyi Yang [Details]

- Diffed human-reviewed labels against the raw model output behind a 26,654-record ESG disclosure corpus (37 ASX entities, 53 report-years, AASB S1/S2, GRI, SASB): review adds evidence conditions 4.38x more often than it removes them (5,310 vs 1,211), concentrated on organisational boundary.
- Showed evidence presence alone reproduces the corpus's disclosed-vs-not labels at 99.92%, so three-class macro-F1 overstates performance and the benchmark endpoint must be criterion-level.
- Built a reproducible pipeline with a 13-issue quality register and scored 29 AASB S2 greenhouse-gas requirements on label balance to select the benchmark subset. Corpus by Zherui Wang (UNSW CSE), used with permission.

Carbon Chain – Cross-Border Sustainability Compliance Verification
China International College Students' Innovation Competition 2026 | Jul. 2026 - Present
Built on the EulerESG platform (in preparation)

- Lead a six-member backend, frontend and evaluation team preparing a competition entry, with planned extension to Chinese disclosure rules.

Existing analysis of the Chinese market

- Designed and implemented a four-layer graph (ESG entities, claim-evidence, provenance, temporal) that scores how far a disclosure claim is supported by traceable evidence, counting evidence by independent source rather than by document.
- Wrote a crawler for the CNINFO portal, collecting 48 sustainability and annual reports from 10 A-share and H-share issuers (2021-2025) as a pilot corpus for the Chinese market.
- Ran a first end-to-end pilot on CATL's carbon neutrality chapter, identifying restated prior-year figures and competing national accounting standards as the main obstacles to cross-issuer comparison.

The Impact of Corporate Environmental, Social and Governance (ESG) Disclosure on Corporate Investment Efficiency
Final Year Project submitted to BNBU [PDF] [Code]
Supervisor: Assoc. Prof. Man Hung Alvin Cheng [Details] | Feb. 2024 - Apr. 2025

- Examined whether ESG disclosure improves investment efficiency through information transparency, using a panel of 19,103 firm-year observations of Chinese A-share listed firms (2013-2023, CSMAR and Sino-Securities).

- Constructed three residual-based investment-efficiency measures (Richardson 2006; Biddle et al. 2009; Chen et al. 2011) and a PCA-based information-transparency index from three daily-frequency measures (liquidity ratio, illiquidity ratio, return reversal), estimated with year and industry fixed effects and firm-clustered standard errors.
- Addressed endogeneity with 2SLS using three peer-average instruments (Kleibergen-Paap Wald F = 4,886) and two-step system GMM: ESG disclosure is positive and significant in the baseline (1% for IE1, 5% for IE2 and IE3) and at the 1% level across all three measures under 2SLS, while the system GMM coefficient is positive but insignificant, indicating that identification rests on cross-firm rather than within-firm variation.
- Established partial mediation through one-year-lagged transparency by 1,000-replication bootstrap (indirect effect $p = 0.022$, identified for the most comprehensive measure); across five heterogeneity dimensions the direct effect holds broadly while the transparency channel concentrates in growth-stage, non-SOE, and high-technology firms.

SELECTED PROJECTS

ESG-Augmented Corporate Credit Risk Prediction and Explainability Analysis
2026, UNSW, Sydney [Code]

- Built an XGBoost pipeline predicting KMV distance-to-default on a 42,000+ firm-year A-share panel (2015-2025) with 18 financial, market and ESG features and RandomizedSearchCV tuning, reaching $R^2 = 0.68$, plus a distress classifier predicting next-year ST/*ST status at ROC-AUC = 0.92; avoided target leakage by rejecting Z-score in favor of a market-based risk measure.
- Quantified the incremental value of ESG through a firm-grouped cross-validation ablation, isolating a small but statistically significant gain (-1.25% RMSE, $p = 0.003$) that held across three distance-to-default definitions and an independent distress target, with multi-layer SHAP explainability over the final model.

Multi-Factor Quantitative Stock Selection & Backtesting
2026, UNSW, Sydney [Code]

- Built a point-in-time factor-research pipeline on 498,893 stock-months across 5,546 A-share firms including delisted names (2015-2025), engineering 8 style and risk factors with cross-sectional winsorization, industry and size neutralization, and a statutory-disclosure lag to remove look-ahead bias.
- Evaluated factor validity via Spearman rank IC, ICIR and factor-decay curves, surfacing a dominant negative turnover effect, small-cap and low-volatility premia, and statistically absent momentum; a walk-forward LightGBM composite reached ICIR 0.72 and a gross long–short quintile Sharpe of 0.93 versus 0.31 for the CSI 300, before transaction costs and A-share short-sale constraints.

Chinese White Dolphin (Sousa chinensis) Habitat Loss and Conservation in Western Hong Kong
2023, BNBU [Code]
Excellent Student of Environmental Awareness

- Built a reproducible PDF-to-panel pipeline (PyMuPDF, pandas) parsing ten years of AFCD marine-mammal monitoring reports into a validated area × monitoring-period panel of 2,209 sightings and 7,023 survey-effort records, aligned with EPD station-level water-quality data.
- Applied a Negative Binomial GLM with area fixed effects and a survey-effort offset, kernel utilization distributions, a Poisson tensor-smooth spatial GAM, and Pettitt / Mann-Kendall change-point tests, estimating an effort-controlled decline of ~12.4% per year (IRR = 0.876) and a 50% core-use area contracting from 45 to 26 km^2 with a 6.5 km south-westward shift.

INDUSTRY EXPERIENCE

Financial Data & Audit Analytics Intern — **Jun. 2024 - Sep. 2024**
Jinchuan Group Corporation Limited — Jinchang, Gansu

- Extracted and reconciled structured financial data from accounting ledgers and vouchers across multiple project entities, performing completeness and accuracy validation in accordance with CSA standards.
- Built Python exception-flagging scripts to cross-validate data across entities and analysed procurement and bidding datasets for cost anomalies, supporting audit risk prioritization and reducing undetected discrepancies in interim audit reports.

Quantitative Cost & Investment Modeling Intern — **Jun. 2022 - Sep. 2022**
State Grid Corporation of China — Jinchang, Gansu

- Built DCF valuation models with scenario and sensitivity analysis (WACC, load growth, residual value) to evaluate NPV and IRR for substation infrastructure investment decisions.
- Developed Excel VBA programs for automated multi-category budget allocation and variance tracking and restructured the cost-investment classification framework to support management reporting on capex/opex distributions.

Investment Data Research Intern **Jul. 2021 - Aug. 2021**

Gansu Electric Power Investment Group Corporation Limited Lanzhou, Gansu

- Conducted sector analysis on China's power market, synthesizing policy signals and market trends into quantitative investment scenarios for the Energy Investment Feasibility Study Report.
- Compiled and standardized multi-source market and financial datasets across subsidiary projects using Python and Excel and built monitoring models for portfolio return and risk metrics (IRR, risk-adjusted indicators) to inform capital allocation.

HONORS & AWARDS

Dean's List BNBU, 2022-2023, 2023-2024, 2024-2025

Bronze Award of Servant Leadership Student Affairs Office, BNBU, 2024

Excellent Student of Environmental Awareness Whole Person Education, BNBU, 2023

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Stata, MATLAB, SAS, SPSS, Excel VBA
- **Research software engineering:** pytest, deterministic seeding, provenance tracking, reproducibility verification, PDF parsing and OCR-assisted extraction, Git Knowledge representation and retrieval (RDF/OWL, SHACL, PROV-O), knowledge graphs, RAG, LLM-as-judge evaluation
- **Machine Learning and statistics:** PyTorch, scikit-learn, XGBoost, SHAP; panel fixed effects, 2SLS, system GMM, bootstrap inference
- **Research Data:** CSMAR, Wind, Bloomberg, WRDS, TuShare, SSE/SZSE/HKEX filings, ESG and sustainability reports
- **Languages:** Mandarin Chinese (native), English (proficient)